

TexEdge: Towards an Approach for Optimization of Sensor Layouts in Textile Interfaces

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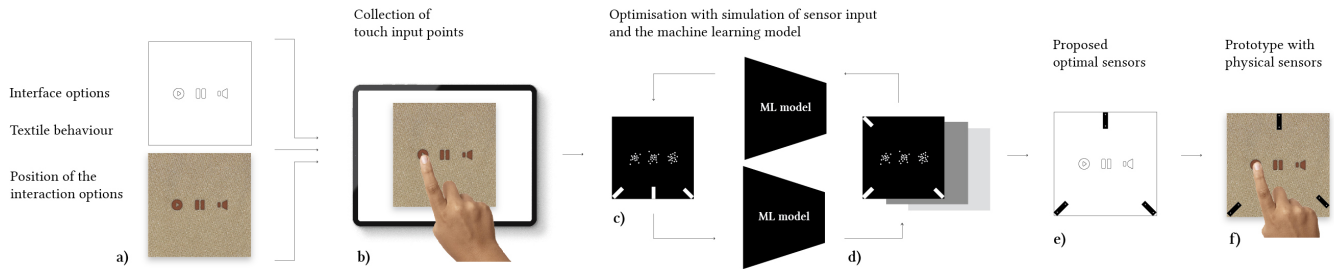


Figure 1: Our *TexEdge* approach proposes the following: A designer starts with exploring user interface options (a) for a flexible textile surface capable of recognizing user gestures (touching, swiping, or other input forms) by mapping them to interactions for a specific application (e.g., controlling a music player). Users perform the intended interactions in a prototypical interface (b), with the collected data serving as a basis for simulations that calculate the strain and stretch enforced on a physical textile interface. The simulated sensor measurements (c) are used as input for machine-learning classification (d). In the training process, the sensor simulation runs simultaneously with the classification to optimize the weights of the machine learning model and the sensor placement to determine an optimal sensor arrangement (e) for the final physical prototype (f).

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1 INTRODUCTION

The smart textiles field is rapidly evolving, using flexible interfaces to enhance user interactions [Pyo et al. 2021]. However, challenges remain, notably in touch input recognition where capacitive or resistive sensors disrupt the textile’s aesthetics and feel [Pointner et al. 2020]. Designing these sensors involves complex considerations like size, positioning, and conductivity [Mlakar et al. 2021], which limits flexibility from a design perspective. Addressing this challenge, our recent work [Zuehlke et al. 2023] demonstrated a method for touch detection on semi-elastic textiles by placing stretch sensors along the edges, which eliminates the necessity of placing the

sensors directly on the surface of the textile, thereby preserving its original texture. To validate it, a textile touchpad prototype was constructed and combined with machine-learning classification (i.e., neural network) of the sensor readings.

While the results were promising, they also highlighted another significant challenge: creating more complex sensor arrays demands more materials and resources, making scalability difficult. The goal of our ongoing work is therefore to strike a balance between minimizing the number of sensors and maintaining the quality of the tracking results by extending this edge-based sensor concept. We propose to do so by introducing a sensor placement optimization approach, which additionally puts a focus on improved integration of user data in interface design (see Figure 1). As a first step towards this, we show first experiments with a scaled-up prototype indicating the feasibility of this approach (see Figure 3).

2 EXPERIMENTS

In [Zuehlke et al. 2023], the edge-based sensing concept was demonstrated with a touchpad prototype (see Figure 2a), where touch gestures on the flexible textile cause deformations of the surface, which are measured by the twelve conductive fabric sensors arranged along the edges of the touch surface. In our current work, we consider a considerably larger setup that may be used in interactive installations. Specifically, it included a 90×90 cm canvas mounted vertically inside a frame and eight stretch sensors (i.e., mechanical load cells) attached to the edges of the fabric (see Figure 2b).

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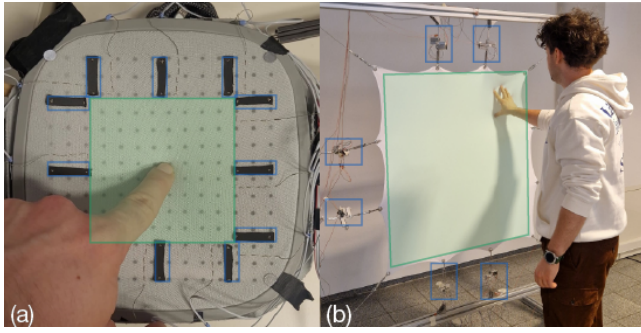


Figure 2: TexEdge prototypes with 8×8 cm (a) and 90×90 cm (b) utilizing twelve textile [Zuehlke et al. 2023] resp. eight mechanical sensors. The interaction area is indicated in green, while blue rectangles indicate the sensor elements.

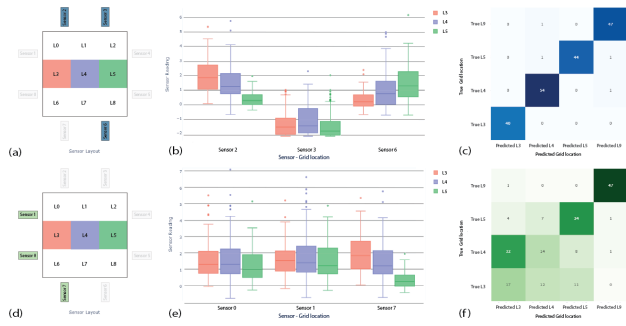


Figure 3: Classification results for two exemplary layouts with three of the eight sensors in the 90×90 cm prototype. Top: optimal sensor layout, utilizing sensors S2, S3, and S6, (a) with clear differences in the distribution of the measurements of the stretch sensors (b) resulting in a high accuracy of 97 % (c) for classifying three locations, L3, L4 and L5. Bottom: non-optimal sensor layout, using sensors S0, S1, and S7, with overlaps in the distribution of the sensor measurements (e) and low accuracy of 59 % (f).

In a data collection experiment, $N = 8$ participants were asked to perform 60 consecutive tap interactions in nine predefined locations, which were individually indicated by back-projection onto the textile surface in randomized order. After cleaning the data, we obtained approximately 150 touch input points per location (L0-L8). The interactions were recorded along with the corresponding readings of the stretch sensors (S0-S7), and served as training data in a subsequent machine-learning optimization process (ten classes: touch locations L0-L8; no interaction, L9). The training data was split randomly into train and validation (30 %) datasets and optimized a Random Forest Classifier with eight inputs, which predicts the interactions with an accuracy of 98.6 %. Further data simulation and classification experiments, where we manually simulated the diverse combinations of the sensors, showed that as little as three of the eight sensors may already be enough to classify touches in three predefined locations (four classes: touch locations L3-L5; no interaction, L9) at sufficient accuracy (see Figure 3).

3 TAKEAWAY

In summary, our experiments demonstrate the feasibility of our approach for the optimization of sensor layouts in textile interfaces. With experiments validating the feasibility of simulating and optimizing sensor arrangements for edge-based sensing in textile interfaces, we propose our *TexEdge* approach (see Figure 1), consisting of the following components:

Design System: The design system allows designers to define interface options under consideration of various parameters such as the size of the canvas, the position and size of touch area and touch point (e.g., one finger touch vs. whole palm press), the depth of the touch, the materiality of the textile, or the complexity of the performed gestures. Based on that, a next step is the collection of real user input in a preliminary (digital) prototype.

Simulation System: The recorded user input is used to simulate deformations of a flexible surface and to calculate corresponding strain and stretch measured by the edge-based sensors. For surface simulation, a bi-linear model only slightly deviates from real textile surfaces, as evaluated in [Zuehlke et al. 2023], and allows the computation of gradients. The number of sensors and their placement are input parameters of the simulation, while the length contractions of the sensors are the output parameters that directly feed a machine learning model for classification of user gestures.

Optimization System: An essential part that we plan to add next will be the application of gradient-based optimization to modify the sensor placement using the simulation of flexible surfaces and edge-based sensors in conjunction with user-gesture classification. After the outlined optimization steps, a material-conserving textile user interface can be manufactured.

For future work, we want to validate and generalize our optimization approach for the design of textile user interfaces. From an interaction perspective, we will include a wider variety of user interface elements and gestures, such as continuous swiping and simple shape drawing. From an implementation perspective, we will focus on refining our approach by collecting and analyzing user interactions to fine-tune our simulation and classifiers (e.g., by using symbolic regression, random forests or neural networks) and develop an optimization algorithm that can dynamically adapt to interface designs and user-input simulations. Finally, we will work on transferring our simulation-trained models to actual textile sensors, considering aspects such as calibration, manufacturing, and sensor accuracy.

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